



02 Student Do: Identifying Misconfigurations (0:15)

In this activity, you will explore a pre-configured Active Directory Certificate Services (AD CS) environment to identify potential misconfigurations that could be exploited by attackers. Follow the steps below:

1. Access the Lab Environment:

- Log into the provided Windows domain environment where AD CS is set up.
- Ensure you have access to the domain controller (DC) and the Certificate Authority (CA) management tools.

2. Examine Certificate Templates:

- Open the Certification Authority management console on the CA server.
- Navigate to the "Certificate Templates" section and review the available templates.
- Look for templates with the following potential issues:
 - **Weak Key Lengths:** Check if any templates allow the issuance of certificates with key lengths less than 2048 bits.
 - **Overly Permissive Permissions:** Identify templates that allow any authenticated user to enroll or request certificates.
 - **Misconfigured Subject Name Formats:** Determine if any templates permit users to specify their own subject names or use wildcards.

3. Review Enrollment Permissions:

- For each certificate template, check the "Security" tab to see which users or groups have enrollment rights.
- Pay attention to templates where permissions are granted too broadly, such as allowing "Domain Users" or "Authenticated Users" to enroll certificates.

4. Investigate CA Configuration:

- Review the CA's general configuration settings, focusing on the following:
 - **CRL Distribution Points:** Verify that the CA's Certificate Revocation List (CRL) distribution points are properly configured and accessible.
 - **Validity Periods:** Check if certificate validity periods are set appropriately; overly long validity periods can increase risk.

5. Document Your Findings:

- As you identify potential misconfigurations, document them along with the associated risks. Include screenshots and detailed notes.
- Think about how each misconfiguration could be exploited to escalate privileges or compromise domain security.

Objective: By the end of this activity, you should be able to identify common misconfigurations in AD CS that could be exploited by an attacker, setting the stage for further exploitation activities.